



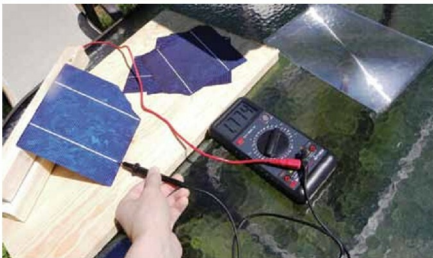
Are You In Suspense?

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Multimeter testing of solar cells

Reference cells consist of a readout device and a $2 \times 2 \text{ cm}^2$ calibrated solar cell made of mono-crystalline silicon. It reads solar simulator irradiance in 'sun' units; one sun is equal to 1000 watts per metre square at 25°C and air mass 1.5 (global reference). The window's material is determined by the type of cell being tested, to reduce spectral mismatch.

Solar simulators. Solar simulation system, or sun simulator, reproduces full spectrum light equal to natural sunlight and provides controlled environment to measure solar cell's performance. Solar simulators expose solar cells to light in a testing environment with a controlled spectrum and perfect lab conditions. This provides a true power-output reading for solar cells.

The simulator consists of a light source and a power supply. The power supply unit provides constant electrical power to the xenon arc lamp. The radiation from the lamp is focused onto an optical integrator that helps produce a uniform diverging beam. The output is a uniform beam that closely matches the sun's radiation spectra for a given air mass.

Solar simulators have following applications for measuring of photovoltaic cell's performance:

1. Determining electrical performance of PV cells
2. Comparing cell's characteristics

among group of cells or different cell designs

3. Repeated measurement of the same cell to study life-cycle performance changes

4. Steady-state testing of PV modules

Portable/handheld measuring instruments

Portable/handheld instruments are required for field measurements.

Multimeter. It is a handheld measuring instrument for ordinary and scheduled maintenance of photovoltaic systems. It is a multi-function electric test device that carries out field measurement of the I-V characteristic—both of a single module and module strings. The instrument measures the I-V characteristic, temperature and incident irradiation of the device being tested. The acquired data is then processed to extrapolate the I-V characteristic at standard test conditions (STC) to compare with the nominal data declared by the modules' manufacturer.

Irradiance meter. Handheld solar radiation meters are used when testing the output of a panel. This device measures the watts per metre square of light. ●

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